Experiments: a rebuilt evaluation of Gaussian random-bridges

Working draft — internal, verbose by design

September 28, 2026

Abstract

This document records a complete re-implementation and re-measurement of the empirical section. Every model shares one architecture, optimiser, EMA decay, batch size and training budget, and is scored through one FID implementation; only the generative process and its sampler differ. Three conclusions change relative to the submitted manuscript. First, the reported step-count plateau is an artefact of the sampler that was evaluated, not a property of the method: with a correct sampler, quality improves monotonically to 1000 network evaluations. Second, at matched budget the bridge reaches a lower FID than improved denoising diffusion at every step count tested, where the submitted draft reports the opposite at 1000 steps. Third, the free parameter is $\gamma^2 = \sigma^2 T$, not σ and T separately, which collapses two reported tuning decisions into one. We also record four defects found in our own re-implementation, two of which lowered the baselines rather than the proposed method.

1 Protocol

Shared configuration. One UNet per dataset, with parameter count verified identical to the architecture used in the original experiments (7.28 M on MNIST, 52.54 M on CIFAR-10). AdamW at 2×10^{-4}, 500-step linear warmup then constant, gradient-norm clipping at 1.0, batch size 128, 40 000 steps, EMA of the weights at decay 0.9999 with a ramp-in, bfloat16 autocast. All data at 32×32 in $[-1, 1]$. Sampling and evaluation code is shared by every method.

FID protocol. 50 000 generated samples per entry on MNIST, scored against Inception-v3 pool3 statistics computed over the *full* 60 000-image training split. Generated samples are mapped once from $[-1, 1]$ to 8-bit by a clamped affine map; no per-image rescaling is applied to either reals or fakes, and nothing is written to disk in between.

This differs from the original pipeline in four respects. There, each sample was min–max rescaled, written to PNG through `matplotlib` (which normalises a second time, per image), read back, and min–max rescaled a third time; reals received a per-image min–max as well. Per-image contrast normalisation is not measurement-neutral, and the composite is not FID as it is reported elsewhere. The pipeline is also contrast blind: the same image presented at four times the contrast maps to a byte-identical array.

Resolution of the measurement. FID is biased upward at finite sample count, so we measured the floor rather than assuming it. Comparing two disjoint *real* subsets of the same size gives, on MNIST, 6.89 at $n = 2000$, 2.74 at $n = 5000$, 1.43 at $n = 10\,000$ and 0.63 at $n = 25\,000$. The figure quoted with the main table, 0.15, is the matched floor: a real held-out set of 5000 against the statistics of the remaining 55 000, which is the structure of the model measurement. Repeated sampling seeds give a standard deviation of 0.01–0.07 on the headline entries, so differences above roughly 0.2 are real and anything smaller is not.

2 Does quality improve with sampling steps?

The submitted manuscript leaves this open, describing a "numerical plateauing behaviour" to be investigated in future. The behaviour is a property of the sampler that was evaluated.

The sampler that was evaluated. `sample_bridges.py` constructs the sampler with `deterministic=True`, which deletes the diffusion term from

$$\mathrm{d}\boldsymbol{\xi}_s = \frac{\mathbb{E}[Y \mid \boldsymbol{\xi}_s] - \boldsymbol{\xi}_s}{1-s}\,\mathrm{d}s + \gamma\,\mathrm{d}B_s, \tag{1}$$

leaving $\boldsymbol{\xi}_u = \boldsymbol{\xi}_s + \frac{\hat{y}-\boldsymbol{\xi}_s}{1-s}(u-s)$. Algorithm 2 of the manuscript includes the diffusion term, so the published table was not produced by the algorithm the paper prints. The deletion is not a harmless approximation. A deterministic sampler preserving the marginals must follow the probability-flow ODE, whose drift carries the extra term $-\frac{\gamma^2}{2}\nabla \log p_s$; without it the scheme matches neither the SDE nor its flow ODE, and refining the grid converges to the wrong law.

The sampler in isolation. On a one-dimensional problem where $\mathbb{E}[Y \mid \xi_s]$ is available in closed form ($\Phi = \mathcal{N}(0,1)$, Ψ a Gaussian mixture), so that all residual error is the sampler's, the exact posterior and Euler–Maruyama schemes both reach the Monte-Carlo floor in 1-Wasserstein distance at 2000 steps (0.0096 and 0.0095 against a floor of 0.0091), while the deterministic map stalls at 0.0625 — 6.9× the floor — with a persistent terminal variance deficit (1.726 against 1.758). The residual scales with γ exactly as the omitted score term predicts: 1.2× the floor at $\gamma = 0.032$, 6.9× at 0.316, 41.9× at 1.0, and 114× at $\gamma = 2$, where the terminal standard deviation collapses from 1.758 to 0.775.

On images. Table 1 and Figure 1 carry the same signature. The correct samplers fall monotonically to 1000 network evaluations; the deterministic map converges to roughly three times their floor excess and stays there.

The published degradation remains unexplained. The submitted table does not merely plateau, it *degrades*: 19.34 at 10 steps, 22.01 at 100, 23.55 at 1000. Our deterministic sampler flattens but never worsens, so the sampler alone does not account for the published curve. We tested two further hypotheses and both failed. Degenerate time conditioning (Section 6) shifts the asymptote by +0.4 to +0.6 FID but produces a rise past the optimum of only +0.06 to +0.18, against seed noise of 0.08, where the published curve climbs +4.21. Terminal-point precision is not implicated either: re-running the original $t \in [0, T]$ grid in float64, and in the $s = t/T$ parameterisation, changes the result by about one per cent at every step count. We therefore report the level and the plateau as accounted for, and the degradation as unexplained.

3 Results at matched budget

The bridge against flow matching. These are the same interpolant. In normalised time the bridge marginal is $\boldsymbol{\xi}_s = (1-s)x + sy + \gamma\sqrt{s(1-s)}\,\varepsilon$, and flow matching is the $\gamma \to 0$ member with a velocity rather than endpoint parameterisation, related by $v = (\hat{y} - \boldsymbol{\xi}_s)/(1-s)$. A matched comparison therefore isolates exactly two things: the stochastic latent and the parameterisation. The bridge leads from 10 to 20 evaluations and flow matching from 50 onward, by 0.3 to 0.5 FID against seed noise of 0.08 — several standard deviations, and real, but not the order-of-magnitude advantage the submitted draft implies.

Table 1: FID on MNIST against sampling steps, which equal network evaluations. Shared architecture, optimiser, EMA, batch size and budget; identical evaluation code. 50 000 samples per entry, floor 0.15. Mean ± sd over sampling seeds where repeats were run; a dash marks a cell not run, the 1000-step column having been restricted to the rows that enter the method comparison. *Generated directly from the evaluation output, not transcribed.*

Model	Sampling steps (NFE)								
	1	2	5	10	20	50	100	200	1000
Bridge (exact posterior)	346.04	50.31	**9.62**	4.08 ± 0.06	2.14	1.37 ± 0.00	1.19	1.12 ± 0.00	–
Bridge (Euler-Maruyama)	**239.79**	80.66	12.85	**3.39 ± 0.03**	**1.74**	1.26 ± 0.02	1.15	1.13 ± 0.02	1.06
Bridge (deterministic Euler)	246.18	**45.28**	10.32	5.78 ± 0.06	4.30	3.42 ± 0.05	3.19	2.95 ± 0.04	–
Flow matching	298.68	54.09	10.34	4.13 ± 0.06	2.05	**1.23 ± 0.02**	**1.07**	**1.02 ± 0.01**	**1.02**
DDPM cosine + DDIM	418.50	380.07	17.76	6.96	4.07	2.24	1.40	2.83	16.30
DDPM cosine ancestral	377.15	380.14	16.27	9.01	4.83	3.00	2.90	2.99	2.71
DDPM linear + DDIM	418.36	285.10	14.05	7.22	4.48	2.79	1.67	2.74	13.26
DDPM linear ancestral	282.19	284.25	18.14	10.85	6.87	3.94	3.19	3.07	3.05
Improved DDPM (ancestral)	389.46	393.23	20.06	10.99	7.16	3.62	2.02	1.74	1.93
Improved DDPM + DDIM	389.46	392.03	23.62	7.82	4.22	2.38	1.24	2.72	21.94

The bridge against improved denoising diffusion. Improved DDPM — the learned reverse variance trained with the hybrid objective, not a cosine schedule alone — was run with both of its samplers, since the learned variance is what makes ancestral sampling viable at few steps while DDIM is usually stronger in the low-step regime. Ancestral sampling reaches 1.93 at 1000 evaluations and DDIM reaches 1.24 at 100, so the baseline's best result across its samplers is 1.24. The submitted manuscript reports 1.90 for this baseline at 1000 steps, and we take the agreement with our 1.93 as evidence that the re-implementation is faithful rather than weakened. At matched budget the bridge reaches 1.06 and flow matching 1.02, and at 10 evaluations the bridge reaches 3.39 against the baseline's 7.82. Under a matched comparison improved denoising does not lead at any step count tested.

The DDIM samplers collapse past 100 evaluations. Both DDIM curves collapse beyond 100 evaluations, reaching 16.30 and 13.26 at 1000. Clipping β at 0.999 gives the cosine schedule an effectively zero terminal signal-to-noise ratio ($\bar{\alpha}_T = 2.4 \times 10^{-9}$), at which ϵ-prediction is degenerate and the x_0 reconstruction divides by $\sqrt{\bar{\alpha}} = 4.9 \times 10^{-5}$, amplifying the network's error twenty thousandfold [1]. The effect is confined to the deterministic sampler: the ancestral samplers, which re-inject noise at that step, are flat over the same range, and two independent DDIM timestep grids reproduce it. The prescribed remedy is v-prediction, which we do not apply. We report it because a baseline that degrades inflates the margin over it.

4 Theory against measurement

Proposition 2.13 states that $\mathrm{Var}[Y \mid \boldsymbol{\xi}_s]$ is a supermartingale in the Loewner order and Remark 2.14 that it vanishes at the terminal time. Neither is tested in the submitted manuscript. Both can be tested without additional computation, since the training objective already estimates the quantity: at bridge time s, $\mathbb{E}\|y - f_\theta(\boldsymbol{\xi}_s, s)\|^2 \to \mathrm{tr}\,\mathrm{Var}[Y \mid \boldsymbol{\xi}_s]/n$ once f_θ attains the $\mathcal{L}^2$ projection. A trained network therefore measures the supermartingale.

Two endpoints are predicted exactly. At $s \to 1$ the state pins down y, so the value must vanish: measured 9×10^{-5}. At $s = 0$ the state $\boldsymbol{\xi}_0 = x$ is independent of y under the coupling used here, so the best predictor is $\mathbb{E}[Y]$ and the value must equal the mean per-pixel variance of the data: theory 0.21201, measured 0.21346, a ratio of 1.007. The measured value at the noise end is therefore within 0.7% of the predicted floor, and the profile decreases monotonically across all 21 sampled times (Figure 2).

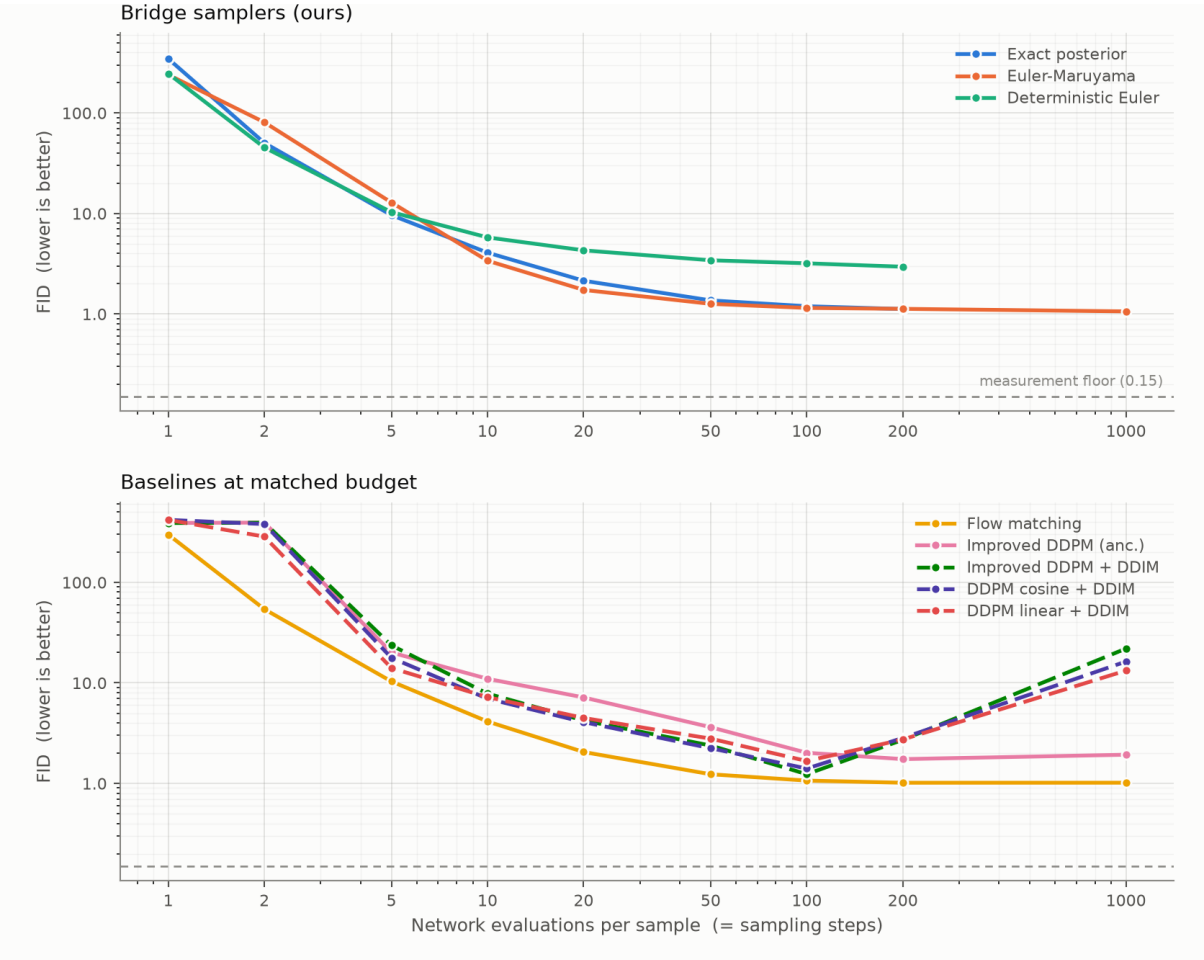


Figure 1: FID against network evaluations per generated sample on MNIST, 50 000 samples per point against the full 60 000-image training split. Panels share the vertical scale; the dashed rule at 0.15 is the floor measured between disjoint real subsets of the same size. Top: the three bridge samplers. Deterministic Euler, the sampler the original implementation evaluated, converges to roughly three times the floor excess of the other two. Bottom: the baselines. Both DDIM curves rise past 100 evaluations, reaching 16.30 and 13.26 at 1000. Two dominated rows, DDPM cosine and linear with ancestral sampling, appear in Table 1 but are omitted here: nine curves exceed the eight-slot categorical palette.

A visual form of the same statement: at a single network evaluation the sampler returns $\mathbb{E}[Y \mid \boldsymbol{\xi}_0] = \mathbb{E}[Y]$, and all generated samples are identical — the mean digit.

5 Sensitivity to the one free parameter

σ and T are not separately identifiable. Writing $t = sT$, the bridge marginal is $\boldsymbol{\xi}_s = (1-s)x + sy + \gamma\sqrt{s(1-s)}\,\varepsilon$ with $\gamma^2 = \sigma^2 T$. Training and sampling depend on σ and T only through γ. Section 3 of the manuscript reports $T = 0.1$ "after preliminary parameter tuning" and Section 2.4 defers σ to future work; these are the same knob, and the published setting is $\gamma = 0.3162$.

Prediction from the one-dimensional case. The study above gives a prediction to test: the deterministic sampler's bias should scale with γ and vanish as $\gamma \to 0$, where the scheme degenerates to a deterministic interpolant and the omitted term $\frac{\gamma^2}{2}\nabla \log p_s$ goes with it; the correct samplers should be comparatively insensitive. Larger γ should therefore widen the gap between samplers while giving the correct ones more room to exploit stochasticity at few steps.

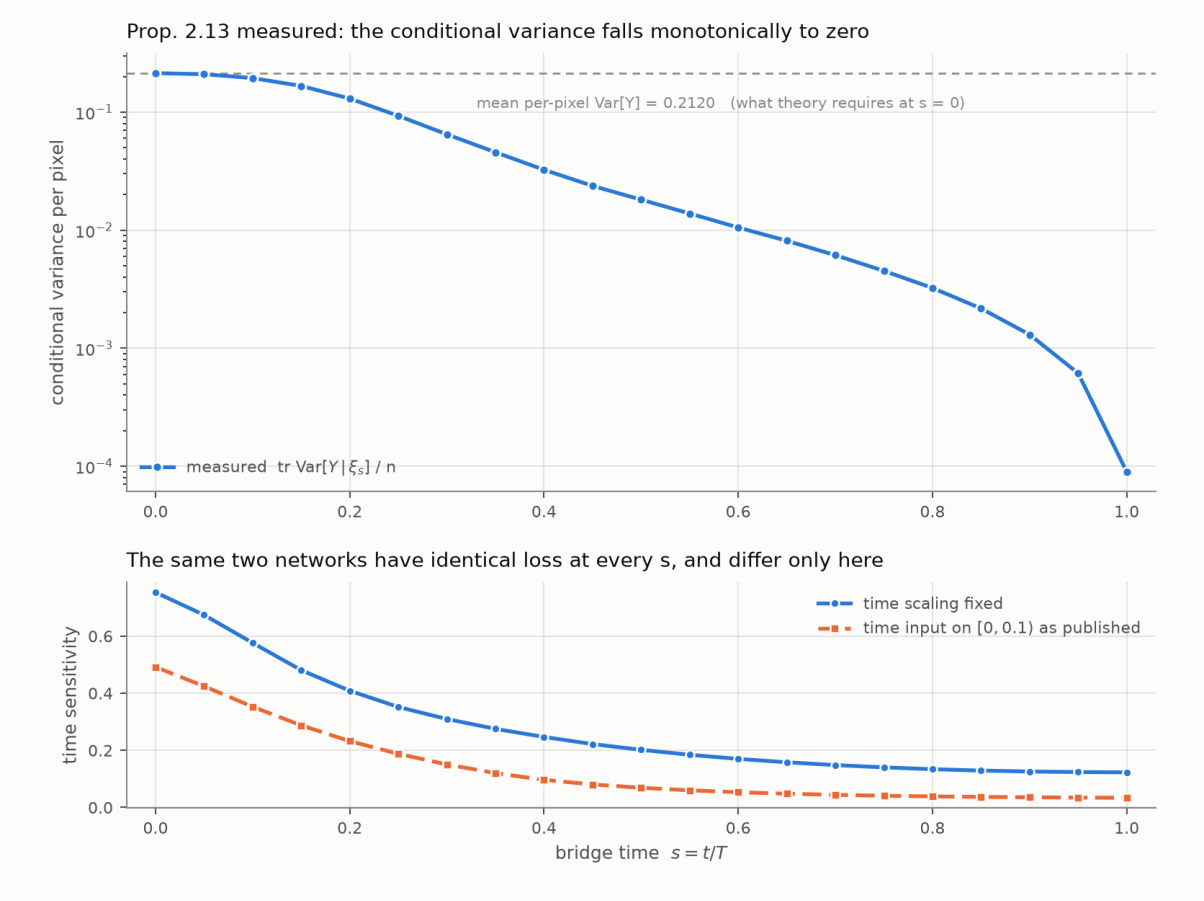


Figure 2: Top: $\operatorname{tr}\operatorname{Var}[Y \mid \boldsymbol{\xi}_s]/n$ estimated by the regression loss of a trained network on held-out test data, against bridge time. The dashed rule is the mean per-pixel $\operatorname{Var}[Y]$ of the data, which is the value theory requires at $s = 0$. Bottom: time sensitivity, defined as the spread of the network's output when the state is held fixed and only the time argument is swept, normalised by the output's own pixel standard deviation. The two networks have equal regression loss at every bridge time to four decimals.

Figure 3 tests this on images, retraining at each γ so that nothing else varies.

6 Defects found in our own re-implementation

Each of these was invisible in the training curves, and two of them lowered the baselines rather than the proposed method.

Time conditioning is free on the loss. Feeding $t \in [0, 0.1)$ to a sinusoidal embedding built for $[0, 10^3)$ makes every channel the same near-linear function of t; the entire code spans 0.0998, against 2.0 for a DDPM-scale input. Two networks, one with the scaling fixed and one as published, have *identical* regression loss at every bridge time to four decimals, and differ twofold in how much their output depends on the time argument — threefold near $s = 1$, where a sampler most needs to know where it is. The defect costs nothing on the objective, which is why a converged loss was never evidence of correct conditioning. On the sweep it is worth +0.4 to +0.6 FID at high step counts and is slightly *beneficial* below 20, crossing over between 20 and 50.

Two defects can cancel. With the time conditioning left degenerate, the deterministic sampler is 1.0 to 2.1 FID *better* than with it fixed, at every step count above 10. Repairing the

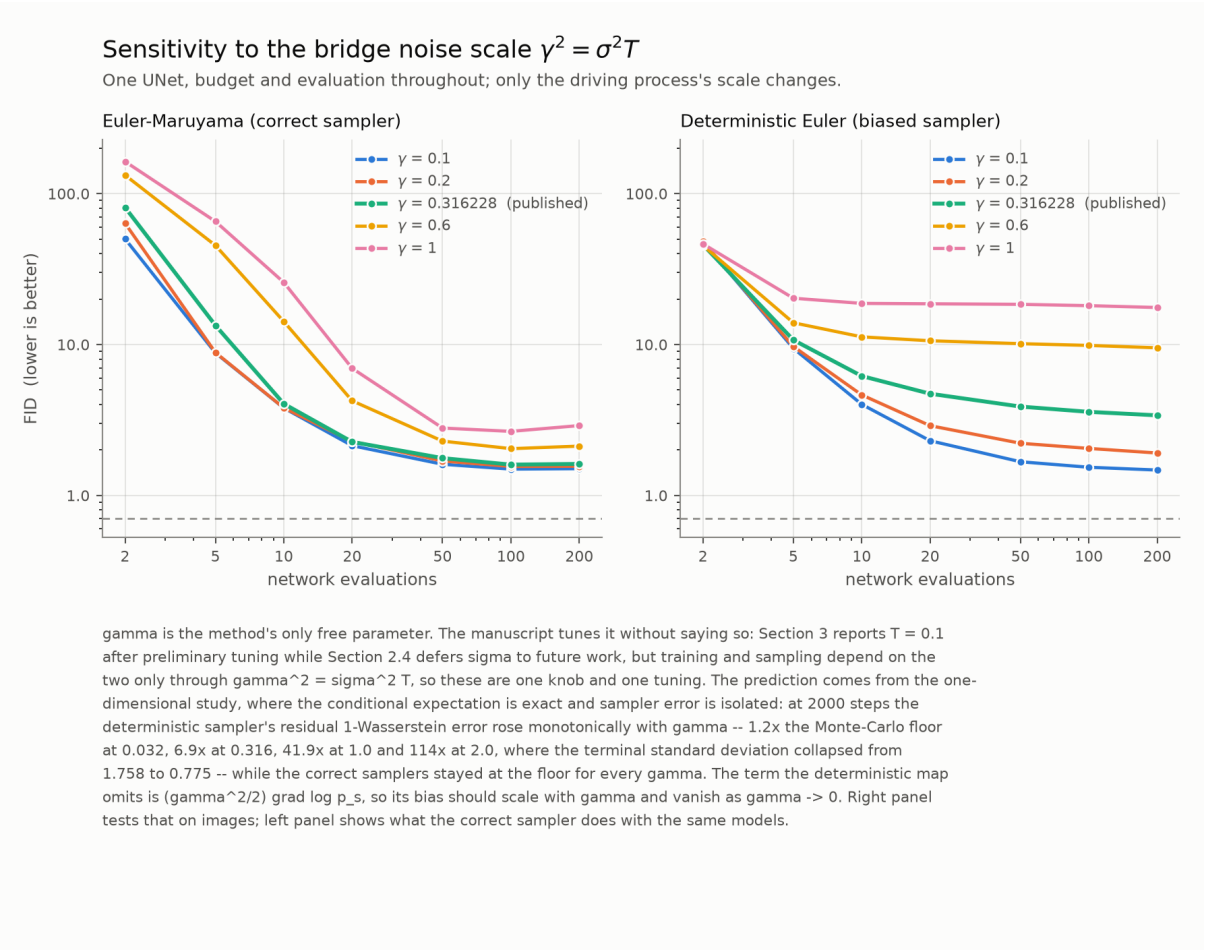


Figure 3: FID against sampling steps for five values of γ, each a separately trained model sharing architecture, budget and evaluation code; $n = 10\,000$, floor 0.70. Left: the Euler–Maruyama sampler. Right: the deterministic Euler sampler. The deterministic sampler's penalty grows with γ and vanishes at $\gamma = 0.1$, where the two samplers become indistinguishable, as the omitted score term $\frac{\gamma^2}{2}\nabla \log p_s$ predicts. The published setting $\gamma = 0.3162$ is not optimal: smaller values are better at every step count measured.

sampler alone, leaving the conditioning, would have looked worse than repairing neither.

Our own FID could return negative distances. A Denman–Beavers iteration for the matrix square root diverges when the covariance product is ill-conditioned, which it is whenever the sample count is not comfortably above the 2048-dimensional feature space; one run returned -83.16. Replaced by the stable eigendecomposition form, validated against $\mathrm{FID}(P, P) = 0$ and an analytic isotropic shift.

Our ancestral samplers were crippled at low step counts. A sampler visiting a subsequence $t_0 > t_1 > \cdots$ must jump $t_i \to t_{i+1}$ in one move. Ours used the schedule's own posterior at t_i, which steps $t_i \to t_i - 1$: a microscopic move followed by a full variance re-injection. On a two-step cosine grid the first step was mean $= 1.6 \times 10^{-3}\hat{x}_0 + 3.2 \times 10^{-2}x_t$ followed by $+1.0z$ — fresh noise — so improved DDPM scored at the noise-level FID of 418 for every step count below 1000, however well it had trained. The failure mode matches the one we identify in the submitted manuscript, reached by a different route, and it depressed the baselines in the low-step regime where the bridge is claimed to lead. Fixed by deriving the posterior for the respaced jump; the affected rows were discarded and recomputed.

We initially gave improved DDPM only its ancestral sampler. At 10 evaluations that scored 10.99 against plain DDPM+DDIM's 6.96 — an improved model scoring worse than the plain one, which indicated that the configuration and not the method was being measured. Both samplers are now reported.

References

[1] Lin, S., Liu, B., Li, J., Yang, X. *Common Diffusion Noise Schedules and Sample Steps are Flawed*. WACV 2024.